GEORGE J. CAZENAVETTE V

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OBJECTIVE *Enter a Computer Science Ph.D. program specializing in artificial intelligence*

EDUCATION

Undergraduate: **Louisiana Tech University**, Ruston, LA
B.S. **Computer Science**, May, 2019
B.S**. Mathematics**, May, 2019
B.S. **Cyber Engineering**, May, 2019
GPA: 4.0,
Enrolled in Honors College

High School: **Saint Paul's School**, Covington, LA
Valedictorian, 2015
GPA: 4.0, National Merit Scholar, ACT: 36
U.S. Dept. of Education Presidential Scholar Semifinalist

ONLINE COURSES
Coursera, Andrew Ng
- Machine Learning
- Deep Learning and Neural Networks
- Structuring Machine Learning Projects
- Hyperparameter Tuning, Regularization, and Optimization
- Convolutional Neural Networks
- Sequence Models

ACTIVITIES AND AWARDS

College of Engineering and Science Outstanding Junior

Association for Computing Machinery, President 2016-2019
- Give presentations and host hackathons
- Teach students at regular study and review sessions

Tau Beta Pi, Upsilon Pi Epsilon, Pi Mu Epsilon
- Engineering, Computing, and Math honor societies
- Tau Beta Pi Scholar (~250 in country)

Institute of Electrical and Electronics Engineers
- IEEEXtreme Programming (top 20 finish in U.S. 2017,18)

Louisiana Tech Jazz Ensemble, Saxophone

SCHOLARSHIPS
Louisiana Tech University
- National Merit Scholarship
- Woodard Endowed Scholarship
- Smith Endowed Scholarship
- Sawyer Endowed Scholarship
- Stiles Endowed Scholarship

Tau Beta Pi
- Record Scholarship

Louisiana Association of Math Educators
- Carol Meyer Memorial Scholarship

CLEARANCE *Secret level obtained September 2017*

RESEARCH AND EXPERIENCE

National Oceanic and Atmospheric Administration,

Senior Capstone Project, Louisiana Tech University, Ruston, LA, Winter 2018-Spring 2019
Advisor: **Dr. Benjamin Drozdenko**
- High-resolution imagery taken from NOAA buoys in the Gulf of Mexico
- Developing Computer Vision algorithms to determine the sea state on the Beaufort Scale
- Also detecting boats and other anomalous objects in the imagery
- Processing must be done on the buoys since transmitting full-size images uses too much power

Senior Honors Research Thesis, Louisiana Tech University, Ruston, LA, Winter 2018-Spring 2019

Advisor: **Dr. Mike O'Neal**
- Researching discrete generation of jazz improvisation over given chords
- Investigating reinforcement learning and generative adversarial network approaches
- Will compile results into an undergraduate thesis and defend before my committee in May 2019

University of Southern California Interaction Lab, Los Angeles, CA, Summer 2018

Advisor: **Dr. Maja Matarić**

- Researched artificial emotion and empathy for socially assistive robotics in the Interaction Lab
- Integrated several platforms to develop a multi-modal learning framework for emotional understanding
- Performed data science and analysis on results of human-robot interaction lab studies using OpenFace
- Developed a heads-up display for Google Glass for interacting with the robot
- Presented poster at the SoCal REU Symposium to students and faculty from the Southern CA area

L3 Technologies, Software Engineering Co-op, Greenville, TX, Summer 2017

- Worked daily with Python, C, and XML on a code generation and translation project
- Internal Research and Development (cannot elaborate further due to ITAR regulations)

Innovative Imaging and Research, Research Intern, Stennis Space Center, MS, Summer 2015, 2016

Advisor: **Dr. Robert Ryan**

- Tested and Developed for NASA's HiDyRS-X project (high dynamic range high-speed video recording)
- Developed software and experiments to characterize Raspberry Pi Camera V2
- Contributed to open-source PiCamera library to increase efficiency of raw image extraction
- Published results in SPIE Journal of Electronic Imaging

PUBLICATION

- Mary A. Pagnutti, Robert E. Ryan, **George J. Cazenavette**, Maxwell J. Gold, Ryan Harlan, Edward Leggett, James F. Pagnutti, "**Laying the foundation to use Raspberry Pi 3 V2 camera module imagery for scientific and engineering purposes**," *J. Electron. Imag*. 26(1) 013014 (11 February 2017)

SKILLS

- **Languages**: Python, Matlab, C#, Java, C, ProLog, Assembly, VHDL, Verilog, SQL
- **Tools**: TensorFlow, Keras, SciKit-Learn, NumPy, Unity, Unreal, Git, LaTeX, Raspberry Pi, Arduino
- **Group Leadership**: Typically a project manager for one or more major group projects each quarter

SELECTED PROJECTS

- Hand-built a neural network using vanilla Java to train on and classify samples from the MNIST data set
- Constructed a two-axis camera system using OpenCV and an Arduino to locate and follow an object
- Built a rudimentary 3D graphics engine from the pixel level using TkInter in Python.
- Developed a short game in Unreal Engine 4 where the player warps between time periods to solve puzzles
- Built a two-factor authentication, Wi-Fi enabled lock box with a touchscreen interface for freshman design
- Wrote conference-style papers to accompany final projects in Data Mining and Distributed Systems courses